Jeongmin Bae

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Professional Summary

I am a Postdoctoral Research Fellow at the University of Melbourne and received my Ph.D. in Artificial Intelligence from Yonsei University in 2026. My research focuses on 3D/4D computer vision, neural rendering, and dynamic scene reconstruction, with particular expertise in Gaussian Splatting and Neural Radiance Field (NeRF)-based methods for dynamic and egocentric scenarios.

I develop learning-based approaches for reconstructing and modeling complex dynamic environments from visual observations, enabling rich 3D representations of real-world scenes. My work has been published at leading computer vision and graphics venues, including ECCV, ICCV, AAAI, and SIGGRAPH.

My current and future research interests include real-time 3D perception and hand-object interaction. I am particularly interested in combining advanced scene reconstruction techniques with foundation models to build richer representations of dynamic environments and enable higher-level perception and understanding.

Research Interests

3D/4D Computer Vision, Neural Rendering, Dynamic Scene Reconstruction, Hand-Object Interaction, Egocentric Vision, Spatial Computing, Generative AI

Technical Skills

3D Vision & Spatial Computing: Gaussian Splatting, Neural Radiance Fields (NeRF), Dynamic Scene Reconstruction, Camera Calibration, LiDAR Processing, Multi-Camera Systems

Machine Learning: PyTorch, Foundation Models, Generative Models

Programming & Tools: Python, Git, Docker, Linux

Publications

- **Hand-4DGS: Feed-Forward 3D Gaussian Splatting for 4D Hand Reconstruction from Egocentric Videos** *Arxiv 2026*
  Jeongmin Bae*, *Seoha Kim*, Marc Pollefeys, Madhi Rad, Youngjung Uh†, Taein Kwon†*
  - This work proposes a generalizable feed-forward Gaussian splatting model that enables hands reconstruction in egocentric videos where existing studies fail. To this end, we utilize hand template priors and learned representations from vision foundation models.
- **4D Scaffold Gaussian Splatting with Dynamic-Aware Anchor Growing for Efficient and High-Fidelity Dynamic Scene Reconstruction** *AAAI 2026*
  Woong Oh Cho, In Cho, Seoha Kim, ***Jeongmin Bae****, Youngjung Uh, Seon Joo Kim*
  - This paper introduces a 4D anchor-based framework for dynamic scene reconstruction that reduces storage costs while maintaining the high visual quality and fast rendering speed of 4D Gaussian methods.
- **Compensating Spatiotemporally Inconsistent Observations for Online Dynamic 3D Gaussian Splatting** *SIGGRAPH 2025*
  Youngsik Yun, ***Jeongmin Bae****, Hyunseung Son, Seoha Kim, Hahyun Lee, Gun Bang, Youngjung Uh*
  - This paper shows that online dynamic scene reconstruction via 3D Gaussians can suffer from temporally inconsistent reconstruction, especially in static regions, depending on the noise in the observations, and addresses this by introducing a residual map.
- **Rethinking Open-Vocabulary Segmentation of Radiance Fields in 3D Space** *AAAI 2025*
  Hyunjee Lee, Youngsik Yun*,* ***Jeongmin Bae****, Seoha Kim, Youngjung Uh*
  - This paper aims to perform clear open-vocabulary semantic segmentation in 3D space and proposes an evaluation protocol that assesses 3D geometry and segmentation simultaneously.
- **Per-Gaussian Embedding-Based Deformation for Deformable 3D Gaussian Splatting** *ECCV 2024*
  Jeongmin Bae*, *Seoha Kim*, Youngsik Yun, Hahyun Lee, Gun Bang, Youngjung Uh*
  - This paper demonstrates that existing deformable 3D Gaussian Splatting models fail to reconstruct complex dynamic scenes and addresses this by replacing the input of the deformation function with learnable embeddings.
- **Optimizing Dynamic NeRF and 3DGS with No Video Synchronization** *ECCV 2024 Wild3D workshop*
  Seoha Kim,* ***Jeongmin Bae****, *Youngsik Yun, Hahyun Lee, Gun Bang, Youngjung Uh*
  - This paper is an extension of Sync-NeRF (Seoha Kim et al., 2024) with the addition of dynamic 3D Gaussian Splatting.

- **Sync-NeRF: Generalizing Dynamic NeRFs to Unsynchronized Videos** *AAAI 2024*
  Seoha Kim, **Jeongmin Bae***, Youngsik Yun, Hahyun Lee, Gun Bang, Youngjung Uh*
  - This paper demonstrates the failure of existing models in dynamic scene reconstruction from unsynchronized multi-view videos and proposes a solution by introducing per-camera time offsets to model calibrated time-dependent representations.
- **BallGAN: 3D-aware Image Synthesis with a Spherical Background** *ICCV 2023*
  *Minjung Shin, Yunji Seo, **Jeongmin Bae**, Young Sun Choi, Hyunsu Kim, Hyeran Byun, Youngjung Uh*
  - This paper aims to learn the 3D geometry of the foreground from a large image collection by modeling the background as a 2D sphere.
- **FurryGAN: High Quality Foreground-aware Image Synthesis** *ECCV 2022*
  ***Jeongmin Bae**, Mingi Kwon, Youngjung Uh*
  - This paper aims to train a GAN to automatically separate fine-detailed foregrounds (such as fur) from large collections of single-view images without using external models or additional supervision.

Experience

- **University of Melbourne – School of Computing and Information Systems []** *Mar. 2026 – Feb. 2027*
  Postdoctoral Research Fellow
  - Postdoctoral Research fellow in the Faculty of Engineering and Information Technology (FEIT), supervised by Prof. Taehyun Rhee.
  - Collaborating on dynamic scene reconstruction and mixed reality projects using multiple 360° panoramic cameras integrated with LiDAR sensors.
- **Electronics and Telecommunications Research Institute (ETRI) []** *Nov. 2022 - Mar. 2025*
  Academic-Research Cooperation
  - Research on enhancing rendering quality in deformable 3D Gaussian splitting for complex dynamic scenes.
  - Research on dynamic scene reconstruction from unsynchronized video datasets using neural radiance fields.
  - Comparison of implicit neural representation models for dynamic scene reconstruction and model reproduction (e.g. DyNeRF, CVPR 2022).

Academic Service

Peer Reviewer: ECCV, ICCV, CVPR, NeurIPS, IEEE TVCG, IEEE TPAMI

Education

- **Yonsei University - Visual Intelligence Lab[]** *Feb. 2022 - Feb. 2026*
  Ph.D. in Artificial Intelligence (Completed), advised by Prof. Youngjung Uh Seoul, South Korea
  - GPA: 4.14/4.3
- **Kookmin University** *Feb. 2013 - Feb. 2021*
  B.S., School of Electronical Engineering Seoul, South Korea
  - Grade: 3.95/4.5 (Major GPA: 4.06/4.5)

Patents

[2025] METHOD AND APPARATUS FOR DECODING IMAGE,
KR 10-2024-0043684

[2023] Method and apparatus for representing dynamic neural radiance fields from unsynchronized videos,
KR 10-2023-0105173

Awards

- **Excellence Award** *2019*
  Python Programming Using Big Data / Samsung Multicampus
  - Received an exceptional excellence award for actively helping and collaborating with others during course completion.
- **Departmental Top Student Scholarship** *2018*
  Kookmin University
  - Won a scholarship in the first semester of my third year as an electronical engineering student, ranked number one out of 222 students in the entire department for the semester.